

4. Group actions

4.1. ACTIONS, G -SETS, AND ORBITS

READING. [Moore, §5.1, pp. 37–41, §8.1–8.2, pp. 125–134].

Definition 4.1 (Permutation). Let X be a set. A *permutation* of X is a map $f : X \rightarrow X$ that is one-to-one and onto.

permutation
置换

The identity map is a permutation, a composite of two permutations is one, and so is an inverse.

Definition 4.2 (Symmetric group). S_X , also written $\text{Sym}(X)$, is the group of permutations of X under composition, $(f_1, f_2) \mapsto f_1 \circ f_2$. For $X = \{1, \dots, n\}$ it is the *symmetric group* S_n , of order $n!$.

symmetric group
对称群

Definition 4.3 (Action). A *left action* of a group G on a set X is a homomorphism

group action
群作用

$$\Phi : G \rightarrow S_X, \quad g \mapsto \Phi(g).$$

Equivalently, it is a map $\varphi : G \times X \rightarrow X$, $\varphi(g, x) = \Phi(g)(x)$, satisfying

$$\varphi(g_1, \varphi(g_2, x)) = \varphi(g_1 g_2, x), \quad \varphi(e, x) = x \quad \text{for all } g_1, g_2 \in G, x \in X.$$

G -set
 G -集合

A set with an action of G is a G -set.

A homomorphism Φ has $\Phi(e) = \text{id}$ and $\Phi(g_1)\Phi(g_2) = \Phi(g_1 g_2)$, which are the two identities for $\varphi(g, x) := \Phi(g)(x)$. Conversely, given φ with the two identities, $\varphi(g, \varphi(g^{-1}, x)) = \varphi(e, x) = x$, so each $\varphi(g, \cdot)$ is a permutation of X with inverse $\varphi(g^{-1}, \cdot)$, and $g \mapsto \varphi(g, \cdot)$ is a homomorphism into S_X . Write

$$g \cdot x := \varphi(g, x), \quad \text{so that} \quad g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x, \quad e \cdot x = x.$$

Sets acted on are taken nonempty.

Example 4.4 (A group acting on itself). Take $X = G$. Left multiplication $g \cdot x := gx$ is an action, since $g_1(g_2 x) = (g_1 g_2)x$ is associativity. Conjugation $g \cdot x := gxg^{-1}$ is an action, since

$$g_1 \cdot (g_2 \cdot x) = g_1(g_2 x g_2^{-1})g_1^{-1} = (g_1 g_2) x (g_1 g_2)^{-1} = (g_1 g_2) \cdot x.$$

right action
右作用

Right multiplication $x \cdot g := xg$ satisfies $(x \cdot g_1) \cdot g_2 = x \cdot (g_1 g_2)$ and defines a *right action*. The corresponding left action is $g \cdot x := xg^{-1}$. For an abelian group conjugation is the trivial action, $gxg^{-1} = x$.

4. Group actions

Example 4.5 (Matrix groups on vectors). $GL(n, \kappa)$ acts on κ^n by $A \cdot \vec{v} := A\vec{v}$, and so does every subgroup of it, for instance $SO(3)$ on $\mathbb{R}^3$.

Example 4.6 (Rotations and translations). Pairs $\{R \mid \vec{\tau}\}$ with $R \in O(3)$ and $\vec{\tau} \in \mathbb{R}^3$ act on $\mathbb{R}^3$ by

$$\{R \mid \vec{\tau}\} \cdot \vec{r} := R\vec{r} + \vec{\tau}.$$

Applying two in succession,

$$\{R_1 \mid \vec{\tau}_1\} \cdot (\{R_2 \mid \vec{\tau}_2\} \cdot \vec{r}) = R_1(R_2\vec{r} + \vec{\tau}_2) + \vec{\tau}_1 = R_1R_2\vec{r} + (R_1\vec{\tau}_2 + \vec{\tau}_1),$$

so the pairs form a group under $\{R_1 \mid \vec{\tau}_1\}\{R_2 \mid \vec{\tau}_2\} = \{R_1R_2 \mid R_1\vec{\tau}_2 + \vec{\tau}_1\}$, the *Euclidean group* $E(3)$, and the formula above is an action of it. Writing a point as the column $(1, \vec{r})^T$, the pair $\{R \mid \vec{\tau}\}$ acts as a 4×4 matrix,

$$\begin{pmatrix} 1 & 0 \\ \vec{\tau} & R \end{pmatrix} \begin{pmatrix} 1 \\ \vec{r} \end{pmatrix} = \begin{pmatrix} 1 \\ R\vec{r} + \vec{\tau} \end{pmatrix},$$

and the group law is matrix multiplication.

Example 4.7 (The square). Let $X = \{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\}$ be the vertices of a square centred at the origin,

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad \vec{v}_3 = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \quad \vec{v}_4 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Realise $D_4 = \langle A, B \mid A^4 = B^2 = (AB)^2 = 1 \rangle$ by matrices as in Example 3.21,

$$\varphi(A) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \varphi(B) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the quarter turn about the origin and the reflection in the horizontal axis. Both carry each vertex to a vertex, so $g \cdot \vec{v} := \varphi(g)\vec{v}$ is an action of D_4 on X . For the quarter turn,

$$(A \cdot \vec{v}_1, A \cdot \vec{v}_2, A \cdot \vec{v}_3, A \cdot \vec{v}_4) = (\vec{v}_2, \vec{v}_3, \vec{v}_4, \vec{v}_1),$$

so A carries X onto itself while moving each vertex to the next. That is the sense in which the quarter turn is a symmetry of the square. The same matrices act on the four edges and on the two diagonals.

Definition 4.8 (Orbit and stabilizer). For $x \in X$, the *orbit* of x and the *stabilizer* of x are

$$G \cdot x := \{g \cdot x : g \in G\} \subset X, \quad G_x := \{g \in G : g \cdot x = x\} \subset G.$$

They are also written $\text{Orb}_G(x)$ and $\text{Stab}_G(x)$.

Proposition 4.9. *The orbits partition X , and G_x is a subgroup of G .*

Euclidean group
欧几里得群

orbit
轨道
stabilizer
稳定化子

Proof. Every $x \in X$ lies in its own orbit, since $x = e \cdot x$, so the orbits cover X . Suppose two of them meet, $y \in (G \cdot x_1) \cap (G \cdot x_2)$, say $y = g_1 \cdot x_1 = g_2 \cdot x_2$. Then $x_1 = (g_1^{-1}g_2) \cdot x_2$, so every point of $G \cdot x_1$ is

$$g \cdot x_1 = (g g_1^{-1} g_2) \cdot x_2 \in G \cdot x_2,$$

giving $G \cdot x_1 \subset G \cdot x_2$, and exchanging the roles of x_1 and x_2 gives the reverse inclusion. Two orbits are therefore equal or disjoint, and X is the disjoint union of its orbits.

For the stabilizer, $e \in G_x$, and if $g, h \in G_x$ then $(gh^{-1}) \cdot x = g \cdot (h^{-1} \cdot x) = g \cdot x = x$, so Lemma 2.5 applies. $\square$

The relation $x \sim y$ when $y = g \cdot x$ for some $g \in G$ is an equivalence relation (Definition 2.9), the orbits are its classes, and the set of orbits is written X/G .

orbit space
轨道空间

Definition 4.10 (Fixed-point set). For $g \in G$, the *fixed-point set* of g is

fixed point
不动点

$$\text{Fix}_X(g) := \{x \in X : g \cdot x = x\} \subset X.$$

So $g \in G_x$ exactly when $x \in \text{Fix}_X(g)$.

Definition 4.11 (Faithful, transitive, free). An action is *faithful*, also called *effective*, if $\Phi : G \rightarrow S_X$ is injective, that is, every $g \neq e$ moves some point, $\text{Fix}_X(g) \neq X$; *transitive* if there is exactly one orbit, that is, any x can be moved to any y ; *free* if every $g \neq e$ moves every point, $\text{Fix}_X(g) = \emptyset$, that is, every stabilizer is trivial.

faithful
忠实
transitive
传递
free
自由

Free implies faithful. The converse fails.

Example 4.12 (The square, continued). On the vertices X of Example 4.7 the action of D_4 is transitive, since the powers of A carry $\vec{v}_1$ to every vertex, and faithful, since a symmetry fixing all four vertices is the identity. It is not free, the stabilizer of $\vec{v}_1$ being $\{e, AB\}$ with AB the reflection in the diagonal through $\vec{v}_1$ and $\vec{v}_3$.

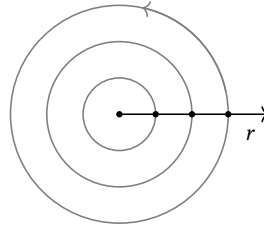
A labelling of the vertices by 1, 2, 3, 4 is an ordered list $(\vec{w}_1, \vec{w}_2, \vec{w}_3, \vec{w}_4)$ of the four vertices, $\vec{w}_j$ being the one labelled j , and there are $4! = 24$ of them. Carrying each label along with its vertex,

$$g \cdot (\vec{w}_1, \vec{w}_2, \vec{w}_3, \vec{w}_4) := (g \cdot \vec{w}_1, g \cdot \vec{w}_2, g \cdot \vec{w}_3, g \cdot \vec{w}_4),$$

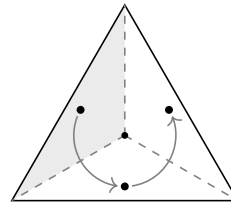
is an action of D_4 on the 24 labellings, and it is free, since a symmetry fixing a labelling fixes all four vertices.

Example 4.13 (Orbits of $SO(2)$ on the plane). $SO(2)$ acting on $\mathbb{R}^2$ has the origin as one orbit and every circle $x^2 + y^2 = r^2$, $r > 0$, as another. The orbit space $\mathbb{R}^2/SO(2)$ is labelled by $r \in [0, \infty)$. The stabilizer of the origin is all of $SO(2)$; the stabilizer of any other point is trivial.

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Example 4.14 (A fundamental domain). The subgroup $C_3 = \{R(0), R(2\pi/3), R(4\pi/3)\}$ of $SO(2)$ acts on the filled equilateral triangle Δ centred at the origin, and every orbit but the centre has three points. The lines from the centre to the vertices cut Δ into three closed thirds, each carried onto the next by $R(2\pi/3)$, so every orbit meets every third. A closed third is a *fundamental domain* for the action. An orbit missing the cuts has one representative in it, while a noncentral orbit on the cuts has two, one on each cut edge, related by $R(2\pi/3)$, and the centre has one. Identifying these boundary pairs gives a model of Δ/C_3 .



Quiz 4.1. List the orbits and a stabilizer for D_4 acting on the four edges of the square, and on its two diagonals. Which of the actions in Example 4.7 are faithful?

4.2. COSETS, LAGRANGE'S THEOREM, AND ORBIT-STABILIZER

Definition 4.15 (Cosets and index). Let H be a subgroup of G . For $g \in G$, the *left coset* and the *right coset* of H through g are

$$gH := \{gh : h \in H\}, \quad Hg := \{hg : h \in H\}.$$

The set of left cosets is written G/H and the set of right cosets $H \backslash G$. The *index* $[G : H]$ is the cardinality of G/H .

The index does not depend on the side, since $gH \mapsto Hg^{-1}$ is a bijection from G/H to $H \backslash G$, with inverse $Hg \mapsto g^{-1}H$. Reversing every product turns each statement about left cosets into the corresponding statement about right cosets, so the choice between the two is a convention, and we take left cosets from here on.

Cosets are orbits. Right multiplication by H is a right action, so by Example 4.4 the left action of H on G carrying the same information is $h \cdot g := gh^{-1}$, and its

fundamental domain
基本域

READING. [Moore, §7.1, pp. 69–72, §8.3, pp. 134–139].

coset
陪集
index
指数

orbits are the left cosets. The left action $h \cdot g := hg$ has the right cosets as its orbits, and the two partitions of G differ in general.

Proposition 4.16. *The left cosets partition G , each of them is in bijection with H , and $g_1H = g_2H$ exactly when $g_2^{-1}g_1 \in H$.*

Proof. The left cosets are the orbits of the action $h \cdot g = gh^{-1}$, so they partition G by Proposition 4.9.

The map $h \mapsto gh$ sends H onto gH by the definition of gH , and it is injective by cancellation, $gh_1 = gh_2$ forcing $h_1 = h_2$. So gH is in bijection with H .

Suppose $g_1H = g_2H$. Since $e \in H$, the element $g_1 = g_1e$ lies in g_1H and hence in g_2H , so $g_1 = g_2h$ for some $h \in H$, and $g_2^{-1}g_1 = h \in H$. Conversely, suppose $g_2^{-1}g_1 = h$ for some $h \in H$. Then $g_1 = g_2h$ lies in g_2H , and $g_1 = g_1e$ lies in g_1H , so the two cosets share the element g_1 and are therefore equal. $\square$

Example 4.17 (Cosets in D_4). In $D_4 = \{e, A, A^2, A^3, B, AB, A^2B, A^3B\}$ take $H = \{e, A, A^2, A^3\}$. Then $eH = H$ and $BH = \{B, BA, BA^2, BA^3\} = \{B, A^3B, A^2B, AB\}$, using $BA = A^{-1}B$. Two cosets, the rotations and the reflections, $[D_4 : H] = 2$. For $H = \{e, B\}$ instead, the cosets are $\{e, B\}$, $\{A, AB\}$, $\{A^2, A^2B\}$, $\{A^3, A^3B\}$, and $[D_4 : H] = 4$.

Lagrange's theorem
拉格朗日定理

Theorem 4.18 (Lagrange). *If G is finite and H is a subgroup, then*

$$|G| = [G : H] |H|.$$

In particular $|H|$ divides $|G|$.

Proof. By Proposition 4.16 the cosets partition G into $[G : H]$ pieces, each in bijection with H and so of size $|H|$. $\square$

Proposition 4.19. *Let G be finite. The order of every element divides $|G|$, and $g^{|G|} = e$ for every g . If $|G|$ is prime then G is cyclic.*

Proof. Let $m = \text{ord } g$. The cyclic subgroup $\langle g \rangle = \{e, g, \dots, g^{m-1}\}$ has m elements, so Theorem 4.18 gives $|G| = km$ for an integer k , and $g^{|G|} = (g^m)^k = e$. If $|G| = p$ is prime and $g \neq e$, then $|\langle g \rangle|$ divides p and is not 1, so $\langle g \rangle = G$. $\square$

Quiz 4.2. Deduce Fermat's little theorem, $a^p \equiv a \pmod{p}$ for a prime p , from Proposition 4.19 applied to the group of Example 3.7.

orbit-stabilizer theorem
轨道-稳定化子定理

4. Group actions

Theorem 4.20 (Orbit-stabilizer). *Let G act on X and let $x \in X$. The map from the orbit $G \cdot x$ to the left cosets of the stabilizer G_x ,*

$$G \cdot x \rightarrow G/G_x, \quad g \cdot x \mapsto gG_x,$$

is well defined and bijective, so $G \cdot x \cong G/G_x$ and

$$|G \cdot x| = [G : G_x].$$

If G is finite, $|G| = |G \cdot x| |G_x|$.

Proof. Well defined. If $g \cdot x = g' \cdot x$ then $(g^{-1}g') \cdot x = x$, so $g^{-1}g' \in G_x$ and $gG_x = g'G_x$. *Onto.* Every coset gG_x is the image of the point $g \cdot x$. *Injective.* If $gG_x = g'G_x$ then $g' = gh$ with $h \in G_x$, so $g' \cdot x = g \cdot (h \cdot x) = g \cdot x$. Since $[G : G_x]$ is the cardinality of G/G_x , the bijection gives $|G \cdot x| = [G : G_x]$, and for finite G Theorem 4.18 turns this into $|G| = |G \cdot x| |G_x|$. $\square$

Example 4.21 (The square, once more). Let D_4 act on the vertices of Example 4.7 and fix $\vec{v}_1$, whose stabilizer is $H = \{e, AB\}$. The four vertices correspond to the four left cosets of H under $g \cdot \vec{v}_1 \mapsto gH$,

vertex	symmetries carrying $\vec{v}_1$ to it
$\vec{v}_1$	$H = \{e, AB\}$
$\vec{v}_2$	$AH = \{A, A^2B\}$
$\vec{v}_3$	$A^2H = \{A^2, A^3B\}$
$\vec{v}_4$	$A^3H = \{A^3, B\}$

Each vertex is reached from $\vec{v}_1$ by exactly two symmetries, the two elements of one coset, and

$$|D_4| = |D_4 \cdot \vec{v}_1| |H| = 4 \times 2 = 8.$$

Example 4.22 ($SO(3)$ on the sphere). $SO(3)$ acts transitively on the unit sphere S^2 , since the proof of Theorem 3.16 rotates $\hat{e}_3$ to any unit vector. The stabilizer of $\hat{e}_3$ is the group of rotations about the third axis, a copy of $SO(2)$. Theorem 4.20 therefore identifies

$$S^2 \longleftrightarrow SO(3)/SO(2), \quad g\hat{e}_3 \mapsto gSO(2),$$

homogeneous space
齐性空间

the sphere with the set of cosets.

A space carrying a transitive action is called a *homogeneous space*, no point being distinguished from any other. Which point is fixed does not matter, since the stabilizers at different points are conjugate,

$$G_{g \cdot x} = g G_x g^{-1},$$

because $h \cdot (g \cdot x) = g \cdot x$ says exactly $g^{-1}hg \in G_x$. On the sphere these are the rotations about the different axes.

Example 4.23 (Wyckoff positions). The square lattice $\mathbb{Z}^2 \subset \mathbb{R}^2$ consists of the points with integer coordinates. Which rigid motions $\{R \mid \vec{t}\}$ of Example 4.6, taken in two dimensions, carry it onto itself? A rotation about the origin does so exactly when its angle is a multiple of $\pi/2$, which gives four, and the reflections in the two axes and in the two diagonals give four more, these eight elements of $O(2)$ being a copy of D_4 . Translation by a lattice vector is a symmetry too, so the full symmetry group

$$G = \{ \{R \mid \vec{t}\} : R \in D_4, \vec{t} \in \mathbb{Z}^2 \}, \quad \{R \mid \vec{t}\} \cdot \vec{r} = R\vec{r} + \vec{t},$$

the plane group $p4mm$, is infinite. As a set it is the direct product $\mathbb{Z}^2 \times D_4$ of Definition 2.11 under $\{R \mid \vec{t}\} \mapsto (\vec{t}, R)$, in which translations would add as $\vec{t}_1 + \vec{t}_2$. Here instead

$$\{R_1 \mid \vec{t}_1\} \{R_2 \mid \vec{t}_2\} = \{R_1 R_2 \mid \vec{t}_1 + R_1 \vec{t}_2\}.$$

Each map $\vec{t} \mapsto R\vec{t}$ is an automorphism of the additive group $\mathbb{Z}^2$, and $(R_1 R_2)\vec{t} = R_1(R_2\vec{t})$, so D_4 acts on $\mathbb{Z}^2$ by automorphisms. A group built from two groups this way is a *semidirect product*, here $G = \mathbb{Z}^2 \rtimes D_4$, and Chapter 6 gives the general construction.

semidirect product
半直积

The *site-symmetry group* of a position $\vec{r}$ is its stabilizer in G ,

site-symmetry group
位置对称群

$$G_{\vec{r}} = \{ \{R \mid \vec{r} - R\vec{r}\} : R \in D_4, \vec{r} - R\vec{r} \in \mathbb{Z}^2 \}.$$

Its orbit $G \cdot \vec{r}$ is infinite, with $|G \cdot \vec{r}| = [G : G_{\vec{r}}]$. Write $[\vec{r}] = \vec{r} + \mathbb{Z}^2$ for a position modulo lattice translations, each class having one representative in the unit cell $[0, 1)^2$. Since $R\mathbb{Z}^2 = \mathbb{Z}^2$, the rule $R \cdot [\vec{r}] = [R\vec{r}]$ is an action of the finite group D_4 on these classes, with stabilizer

$$H_{\vec{r}} = \{ R \in D_4 : R\vec{r} - \vec{r} \in \mathbb{Z}^2 \},$$

and $\{R \mid \vec{r} - R\vec{r}\} \mapsto R$ is an isomorphism $G_{\vec{r}} \cong H_{\vec{r}}$, since R determines the translation and $(\vec{r} - R_1\vec{r}) + R_1(\vec{r} - R_2\vec{r}) = \vec{r} - R_1 R_2\vec{r}$. Modulo lattice translations the orbit $G \cdot \vec{r}$ becomes $D_4 \cdot [\vec{r}]$, whose cardinality is the number of equivalent positions in $[0, 1)^2$, the *multiplicity*,

multiplicity
重数

$$|D_4 \cdot [\vec{r}]| = [D_4 : H_{\vec{r}}] = \frac{8}{|H_{\vec{r}}|} = \frac{8}{|G_{\vec{r}}|}.$$

Take $\vec{r} = (x, 0)$ with $0 < x < \frac{1}{2}$. The identity and the reflection B in the horizontal axis are the two elements satisfying $R\vec{r} - \vec{r} \in \mathbb{Z}^2$, so $G_{\vec{r}} = \{\{1 \mid \vec{0}\}, \{B \mid \vec{0}\}\}$ has order 2, the multiplicity is 4, and the four positions of the unit cell are

$$(x, 0), \quad (1 - x, 0), \quad (0, x), \quad (0, 1 - x).$$

At $\vec{r} = (\frac{1}{2}, \frac{1}{2})$ every $R \in D_4$ satisfies $R\vec{r} - \vec{r} \in \mathbb{Z}^2$, so $G_{\vec{r}} \cong D_4$ and the multiplicity is 1. The operation $\{A \mid (1, 0)\}$ is the counterclockwise quarter turn about this position.

Wyckoff positions
Wyckoff 位置

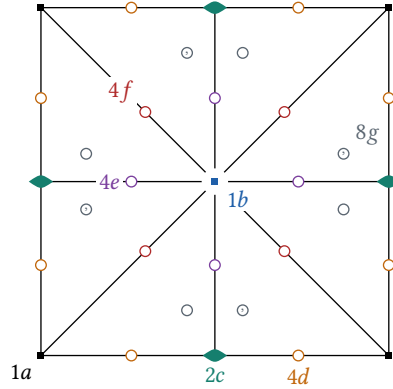
The seven families below are the *Wyckoff positions* of $p4mm$. Take $0 < x < \frac{1}{2}$ in

4. Group actions

the one-parameter rows and $0 < y < x < \frac{1}{2}$ in the last row.

representative	$ G_{\vec{r}} $	multiplicity	label
$(0, 0)$	8	1	$1a$
$(\frac{1}{2}, \frac{1}{2})$	8	1	$1b$
$(\frac{1}{2}, 0)$	4	2	$2c$
$(x, 0)$	2	4	$4d$
$(x, \frac{1}{2})$	2	4	$4e$
(x, x)	2	4	$4f$
(x, y)	1	8	$8g$

The number in each label is the multiplicity. The diagram shows one orbit from each family.



The diagram includes all four edges of $[0, 1]^2$, and corresponding points on opposite edges represent the same translation class, so counting representatives in $[0, 1)^2$ gives one corner position and two edge-centre positions. Mirror lines, rotation centres and orbits are drawn with symbols from the *International Tables* [ITA, §2.1], solid lines marking mirrors, filled squares four-fold centres and filled lenses two-fold centres. In the general orbit $8g$ a comma marks the four points obtained from the other four by an operation with $\det R = -1$.

4.3. EQUIVARIANT MAPS

READING. [Moore, §5.3, pp. 44–45].

equivariant map
等变映射

Definition 4.24 (Equivariant map). Let X and Y be G -sets. A map $f : X \rightarrow Y$ is *equivariant* if

$$f(g \cdot x) = g \cdot f(x) \quad \text{for all } g \in G, x \in X,$$

that is, the square

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ g \cdot \downarrow & & \downarrow g \cdot \\ X & \xrightarrow{f} & Y \end{array}$$

G -space
 G -空间

commutes for every g . A bijective equivariant map is an *isomorphism of G -sets*, and its inverse is then equivariant too.

In topology a G -space is a topological space X with an action of a topological group G for which $G \times X \rightarrow X$ is continuous, and a morphism of G -spaces is a continuous equivariant map. Some authors use the word for a G -set with no topology in sight.

The bijection of Theorem 4.20 is an isomorphism of G -sets. Write ψ for it, $\psi(g \cdot x) = gG_x$. The coset set G/G_x carries the action $a \cdot (gG_x) := (ag)G_x$, well defined since $gG_x = g'G_x$ gives $(ag)G_x = (ag')G_x$, and

$$\psi(a \cdot (g \cdot x)) = \psi((ag) \cdot x) = (ag)G_x = a \cdot (gG_x) = a \cdot \psi(g \cdot x),$$

so the square

$$\begin{array}{ccc} G \cdot x & \xrightarrow{\psi} & G/G_x \\ a \cdot \downarrow & & \downarrow a \cdot \\ G \cdot x & \xrightarrow{\psi} & G/G_x \end{array}$$

commutes for every $a \in G$. So $G \cdot x \cong G/G_x$ as G -sets, and every transitive G -set is a coset space G/H , with H the stabilizer of any of its points. Example 4.22 is the case $S^2 \cong SO(3)/SO(2)$.

Example 4.25 (Equivariant maps of the line). Let $\mathbb{Z}$ act on $\mathbb{R}$ by $n \cdot x := x + n$. The orbits are the classes $x + \mathbb{Z}$, one through each point of $[0, 1)$, so $\mathbb{R}/\mathbb{Z}$ is a circle. Give a map $f : \mathbb{R} \rightarrow \mathbb{R}$ this action on its domain and on its codomain. It is equivariant when $f(x + n) = f(x) + n$ for all n , that is, when $P(x) := f(x) - x$ satisfies $P(x + 1) = P(x)$. So the equivariant maps are exactly $f(x) = x + P(x)$ with P unchanged by $x \mapsto x + 1$, the simplest being $f(x) = x + c$. Equivariance under all real translations leaves only these, since setting $x = 0$ in $f(x + t) = f(x) + t$ gives $f(t) = f(0) + t$ for every $t \in \mathbb{R}$.

[Moore, p. 45] is wrong to say that the analytic equivariant maps here are the translations. That is true for the action of $\mathbb{R}$. For $\mathbb{Z}$ the map $x + \sin 2\pi x$ is analytic and equivariant.

Quiz 4.3. D_4 acts on the four vertices of the square and on its four edges. Send each vertex to the edge joining it to the next vertex clockwise. Is this map equivariant? Is there an equivariant bijection from the vertices to the edges at all? Compare stabilizers.

4.4. CONJUGACY, CENTRALIZERS, AND NORMALIZERS

READING. [Moore, §7.2, pp. 72–80].

For the conjugation action of Example 4.4, $g \cdot h = ghg^{-1}$, the orbits and stabilizers have their own names.

Definition 4.26 (Conjugacy class and centralizer). Elements h and ghg^{-1} are *conjugate*. The orbit of h under conjugation is its *conjugacy class*

$$\text{Cl}(h) := \{ ghg^{-1} : g \in G \},$$

conjugate
共轭
conjugacy class
共轭类
centralizer
中心化子

4. Group actions

and its stabilizer is the *centralizer*

$$C_G(h) := \{g \in G : ghg^{-1} = h\} = \{g \in G : gh = hg\}.$$

The classes partition G and the centralizer is a subgroup, by Proposition 4.9, and by Theorem 4.20

$$|\text{Cl}(h)| = [G : C_G(h)], \quad (4.1)$$

so for finite G the size of every conjugacy class divides $|G|$.

The centre $Z(G)$ of Definition 2.15 reappears here. An element z has $\text{Cl}(z) = \{z\}$ exactly when every g commutes with it, that is, when $C_G(z) = G$, so

$$Z(G) = \{h : |\text{Cl}(h)| = 1\} = \bigcap_{h \in G} C_G(h).$$

For any h the centralizer is the largest subgroup in which h is central, $h \in Z(C_G(h))$, since $h \in Z(K)$ forces $K \subset C_G(h)$. For finite G , (4.1) reads $|\text{Cl}(h)| = |G|/|C_G(h)|$, so the larger the centralizer the smaller the class. The same definition serves a subset $H \subset G$: its centralizer $C_G(H) := \{g : gh = hg \text{ for all } h \in H\}$ is the intersection of the $C_G(h)$, hence a subgroup, and $C_G(G) = Z(G)$.

Example 4.27 (Classes of D_4). With $D_4 = \langle A, B \mid A^4 = B^2 = 1, BAB = A^{-1} \rangle$, conjugating A by B gives $BAB^{-1} = A^{-1} = A^3$, and conjugating by a rotation changes nothing, so $\text{Cl}(A) = \{A, A^3\}$. Conjugating B by A gives $ABA^{-1} = A(BA^{-1}) = A(AB) = A^2B$, so $\text{Cl}(B) = \{B, A^2B\}$, and likewise $\text{Cl}(AB) = \{AB, A^3B\}$. Since A^2 commutes with A and with B , it is central. The five classes are

$$\{e\}, \quad \{A^2\}, \quad \{A, A^3\}, \quad \{B, A^2B\}, \quad \{AB, A^3B\},$$

of sizes 1, 1, 2, 2, 2, each dividing 8, and $Z(D_4) = \{e, A^2\}$. The two reflection classes are the reflections in diagonals and the reflections in edge bisectors: conjugation cannot turn one kind into the other.

Example 4.28 (Classes in unitary groups). The spectral theorem writes a unitary u as

$$u = QDQ^\dagger, \quad D = \text{diag}(z_1, \dots, z_n),$$

where the columns of $Q \in U(n)$ are an orthonormal basis of eigenvectors and D holds the corresponding eigenvalues. These have $|z_i| = 1$, since u leaves the length of every vector unchanged and an eigenvector $\vec{v}$ with $u\vec{v} = z_i\vec{v}$ has $\|\vec{v}\| = |z_i| \|\vec{v}\|$.

Let $u, v \in U(n)$ have the same eigenvalues with multiplicity, and order the eigenbasis of v to match, $v = PDP^\dagger$ with the same D . The unitary $PQ^\dagger$ carries the eigenbasis of u to that of v , and

$$v = PDP^\dagger = (PQ^\dagger)QDQ^\dagger(PQ^\dagger)^{-1} = (PQ^\dagger)u(PQ^\dagger)^{-1},$$

and conversely conjugation preserves eigenvalues and multiplicities. So two unitary matrices are conjugate in $U(n)$, and equally in $GL(n, \mathbb{C})$, exactly when

their eigenvalues agree with multiplicity, and a class is labelled by D up to permutation of its diagonal entries.

Every $u \in SU(2)$ has the axis-angle form of Remark 3.17,

$$u(\hat{n}, \theta) = \cos \frac{\theta}{2} \mathbb{1} - i \sin \frac{\theta}{2} \hat{n} \cdot \vec{\sigma}, \quad \hat{n} \in S^2, \quad 0 \leq \theta \leq 2\pi,$$

this being the general element $\begin{pmatrix} z & -w^* \\ w & z^* \end{pmatrix}$ of Example 2.20 written in the Pauli basis, with $\cos \frac{\theta}{2} = \Re z$.

Conjugation turns the axis and leaves the angle alone. Writing R for the map of Definition 3.15, fixed by $g(\vec{x} \cdot \vec{\sigma})g^\dagger = (R(g)\vec{x}) \cdot \vec{\sigma}$,

$$g u(\hat{n}, \theta) g^{-1} = \cos \frac{\theta}{2} \mathbb{1} - i \sin \frac{\theta}{2} (R(g)\hat{n}) \cdot \vec{\sigma} = u(R(g)\hat{n}, \theta).$$

As R is onto $SO(3)$, which carries $\hat{n}$ to every unit vector,

$$\text{Cl}(u(\hat{n}, \theta)) = \{u(\hat{m}, \theta) : \hat{m} \in S^2\},$$

so the class is fixed by θ alone, equivalently by $\text{Tr } u = 2 \cos \frac{\theta}{2}$. At $\theta = 0$ and $\theta = 2\pi$ it is $\{\mathbb{1}\}$ and $\{-\mathbb{1}\}$.

Definition 4.29 (Class function). A function $f : G \rightarrow \mathbb{C}$ with $f(ghg^{-1}) = f(h)$ for all g, h is a *class function*: a function on the set of conjugacy classes.

class function
类函数

On $GL(n, \mathbb{C})$ the trace, the determinant, and every coefficient of the characteristic polynomial $\det(x\mathbb{1} - A)$ are class functions, since $\det(x\mathbb{1} - PAP^{-1}) = \det(P(x\mathbb{1} - A)P^{-1}) = \det(x\mathbb{1} - A)$. On $SU(2)$ the trace alone labels the class.

Subgroups can be conjugated as well as elements. If H is a subgroup then $gHg^{-1} = \{ghg^{-1} : h \in H\}$ contains e and satisfies

$$(gh_1g^{-1})(gh_2g^{-1})^{-1} = g(h_1h_2^{-1})g^{-1} \in gHg^{-1},$$

so it is a subgroup. The rule $g \cdot H := gHg^{-1}$ is an action, since $eHe^{-1} = H$ and $g_1(g_2Hg_2^{-1})g_1^{-1} = (g_1g_2)H(g_1g_2)^{-1}$.

Definition 4.30 (Normalizer). The *normalizer* of a subgroup H is its stabilizer under this action,

normalizer
正规化子

$$N_G(H) := \{g \in G : gHg^{-1} = H\}.$$

It is a subgroup by Proposition 4.9, and it contains H , since $hHh^{-1} = H$ for $h \in H$, as well as $C_G(H)$. Any subgroup $K \subset G$ with $kHk^{-1} = H$ for all $k \in K$ lies in $N_G(H)$, so $N_G(H)$ is the largest subgroup of G in which conjugation carries H onto itself, as $C_G(h)$ is the largest one in which h is central. The number of subgroups conjugate to H is $[G : N_G(H)]$, by Theorem 4.20 once more.

Quiz 4.4. Find the conjugacy classes and the centre of the quaternion group Q of Example 2.25, and check (4.1) for each class.

inner automorphism
内自同构

Conjugation by a fixed a is more than a permutation of G : it respects the multiplication, $(axa^{-1})(aya^{-1}) = a(xy)a^{-1}$, so $I_a : x \mapsto axa^{-1}$ is an automorphism of G , an *inner automorphism*. Since $I_a I_b = I_{ab}$, the map $I : G \rightarrow \text{Aut}(G)$ is a homomorphism, and its kernel is $Z(G)$. Its image is written $\text{Inn}(G)$.

4.5. THE CLASS EQUATION

READING. [Moore, §9.2,
pp. 187–189].

class equation
类方程

Theorem 4.31 (Class equation). *Let G be finite, and let $g_1, \dots, g_r$ be one element from each conjugacy class with more than one element. Then*

$$|G| = |Z(G)| + \sum_{i=1}^r [G : C_G(g_i)],$$

where every term of the sum is greater than 1 and divides $|G|$.

Proof. The classes partition G . The singleton classes are exactly the elements of $Z(G)$, contributing $|Z(G)|$. Each remaining class contributes its size, which is $[G : C_G(g_i)]$ by (4.1), greater than 1 by choice of g_i and a divisor of $|G|$ by Theorem 4.18. $\square$

For D_4 this reads $8 = 2 + 2 + 2 + 2$. The equation is most useful when $|G|$ has few divisors.

p -group
 p -群

A finite group whose order is a power of a prime p is a *p -group*.

Proposition 4.32 (Centre of a p -group). *A p -group of order p^n with $n \geq 1$ has a nontrivial centre.*

Proof. Each term $[G : C_G(g_i)]$ divides p^n and exceeds 1, so it is a positive power of p . Then $|Z(G)| = |G| - \sum_i [G : C_G(g_i)]$ is a multiple of p , and it is at least 1 because $e \in Z(G)$. So $|Z(G)| \geq p$. $\square$

Lagrange gives a necessary condition on the order of a subgroup and not a sufficient one. The twelve rotations of a regular tetrahedron have no subgroup of order 6, and no smaller group fails this way. For a prime divisor a subgroup of that order does always exist, by another orbit count.

Cauchy's theorem
柯西定理

Theorem 4.33 (Cauchy). *Let G be finite. If a prime p divides $|G|$ then G has an element of order p .*

Proof. Let $X = \{(g_1, \dots, g_p) \in G^p : g_1 g_2 \cdots g_p = e\}$. Choosing $g_1, \dots, g_{p-1}$ freely determines g_p , so $|X| = |G|^{p-1}$, a multiple of p . The cyclic group $\mathbb{Z}_p$ acts on X by rotating the entries, $(g_1, \dots, g_p) \mapsto (g_p, g_1, \dots, g_{p-1})$; the product is still e , since $g_p(g_1 \cdots g_{p-1}) = g_p g_p^{-1}$. By Theorem 4.20 every orbit has 1 or p elements, as $|\mathbb{Z}_p| = p$ is prime. A one-element orbit is a tuple fixed by rotation, that is,

$(g, g, \dots, g)$ with $g^p = e$. Writing N_1 for the number of these and N_p for the number of p -element orbits,

$$|G|^{p-1} = N_1 + pN_p,$$

so p divides N_1 . Since $(e, \dots, e)$ is one fixed tuple, $N_1 \geq 1$, hence $N_1 \geq p$, and some $g \neq e$ has $g^p = e$. Its order divides p and is not 1, so it is p . $\square$

4.6. FINITE ROTATION GROUPS

READING. [Artin, §6.12]. Not in Moore.

A finite subgroup of $SO(3)$ acts on the set of unit vectors fixed by its nonidentity rotations. Counting that set with orbit–stabilizer leaves a few possibilities, and identifying which solids realise them gives the classification.

Lemma 4.34. *A rotation $A \in SO(3)$ with $A \neq \mathbb{1}$ fixes exactly two unit vectors, $\pm \hat{n}$ for its axis $\hat{n}$. A finite subgroup of $SO(3)$ all of whose elements fix a given unit vector is cyclic.*

Proof. For an eigenvector $\vec{v} \in \mathbb{C}^3$ with $A\vec{v} = \lambda\vec{v}$, orthogonality gives $A^\dagger A = \mathbb{1}$ and

$$\|\vec{v}\|^2 = \|A\vec{v}\|^2 = |\lambda|^2 \|\vec{v}\|^2,$$

so $|\lambda| = 1$. The characteristic polynomial has real coefficients. If a non-real conjugate pair occurs, its product is 1 and the third eigenvalue is $\det A = 1$. Otherwise all three eigenvalues are ± 1 with product 1, so at least one is $+1$.

Choose a real unit eigenvector $\hat{n}$ with $A\hat{n} = \hat{n}$. Since $A^{-1}\hat{n} = \hat{n}$, we have $\hat{n} \cdot A\vec{v} = \hat{n} \cdot \vec{v}$, so the plane $\hat{n}^\perp$ is invariant, and A restricted to it is orthogonal with determinant 1. For $A \neq \mathbb{1}$ it is a nonidentity rotation of that plane, whose only fixed vector is $\vec{0}$. So the fixed vectors of A are the line $\mathbb{R}\hat{n}$ and its fixed unit vectors are $\pm \hat{n}$.

Let H be a finite subgroup fixing a unit vector $\hat{n}$. Restriction to $\hat{n}^\perp$ identifies H with a finite subgroup K of $SO(2) \cong U(1)$ (Example 3.14). Put $m = |K|$. Every $z \in K$ satisfies $z^m = 1$ by Proposition 4.19, so $K \subset \mu_m$ of Example 2.8. Both have m elements, so $K = \mu_m$ and H is cyclic. $\square$

poles
极点

The two fixed unit vectors of a nonidentity rotation A , the set $\text{Fix}_{S^2}(A)$, are its *poles*.

Theorem 4.35 (Finite subgroups of $SO(3)$). *Up to conjugacy in $SO(3)$, a finite subgroup of $SO(3)$ is one of: a cyclic group C_n of rotations about one axis, $n \geq 1$; a dihedral group D_n of order $2n$, $n \geq 2$; or one of three groups of orders 12, 24, 60, the rotation groups of the regular tetrahedron, of the octahedron (equally of the cube), and of the icosahedron (equally of the dodecahedron).*

4. Group actions

Proof. The trivial group is C_1 . Let $G \subset SO(3)$ be finite with $|G| = N > 1$, and let P be the set of poles of its nonidentity elements. G acts on P : if A fixes p then BAB^{-1} fixes Bp . Count the pairs (A, p) with $A \neq 1$ and p a pole of A in two ways. Each of the $N - 1$ nonidentity rotations has two poles, giving $2(N - 1)$. On the other side, a pole p with stabilizer of order n_p is fixed by the $n_p - 1$ nonidentity elements of G_p , and by Theorem 4.20 its orbit has N/n_p poles, all with stabilizer order n_p . Summing over the orbits $P_1, \dots, P_k$ with stabilizer orders $n_1, \dots, n_k$,

$$2(N - 1) = \sum_{i=1}^k \frac{N}{n_i} (n_i - 1), \quad \text{that is} \quad 2 - \frac{2}{N} = \sum_{i=1}^k \left(1 - \frac{1}{n_i}\right).$$

Each $n_i \geq 2$, so each term on the right is at least $\frac{1}{2}$, while the left side is less than 2: hence $k \leq 3$. And $k = 1$ is impossible, since $1 - 1/n_1 < 1 \leq 2 - 2/N$.

Two orbits. $2/N = 1/n_1 + 1/n_2$ with $n_i \leq N$ forces $n_1 = n_2 = N$: every element of G fixes both poles of one axis, and G is cyclic by Lemma 4.34.

Three orbits. $1 + 2/N = 1/n_1 + 1/n_2 + 1/n_3$ with $n_1 \leq n_2 \leq n_3$. Not all can be 3 or more, so $n_1 = 2$, and then $1/n_2 + 1/n_3 = 1/2 + 2/N > 1/2$ leaves

$$(2, 2, n), N = 2n; \quad (2, 3, 3), N = 12; \quad (2, 3, 4), N = 24; \quad (2, 3, 5), N = 60.$$

Dihedral. In the case $(2, 2, n)$ the orbit with stabilizer order n has two poles p, q . A nonidentity element of G_p fixes q as well, so $q = -p$. Write $G_p = \langle a \rangle$, of order n . Any $b \in G$ carrying p to $-p$ is a half-turn about an axis perpendicular to p , the two cosets of G_p are G_p and bG_p , and $bab^{-1} = a^{-1}$. So G is the D_n of Example 2.24, its planar reflections realised as spatial half-turns.

Polyhedral. In the case $(2, 3, 3)$ take a four-point orbit V with stabilizer order 3. The C_3 fixing $p \in V$ acts on the other three points with orbits of size 1 or 3. They cannot all be fixed, since a nonidentity rotation fixes only two unit vectors, so they form one three-point orbit, and p has the same inner product with each. Transitivity of G on V then makes all six inner products equal, and V is a regular tetrahedron. In the case $(2, 3, 4)$ the six-point orbit V with stabilizer order 4 is the only pole orbit of its size, so it contains $-p$ along with p . The C_4 fixing p cycles the four points other than $\pm p$, which therefore form a square in a plane perpendicular to p , and closure under $p \mapsto -p$ puts that plane through the origin: V is a regular octahedron. In the case $(2, 3, 5)$, in sketch, the unique twelve-point pole orbit V is likewise closed under $p \mapsto -p$, and the C_5 fixing p splits the other ten points into two antipodal regular pentagons at heights $\pm a$, $0 \leq a < 1$. By transitivity every vertex has inner products $-1, a$ five times, and $-a$ five times with the others. Two consecutive vertices of one pentagon have inner product $d = a^2 + (1 - a^2) \cos(2\pi/5)$, and $0 < d < 1$ forces $d = a$, whose admissible root is $a = 1/\sqrt{5}$: V is a regular icosahedron.

Finally G is the full rotation group of the solid in each case. A rotation preserving the solid is determined by the images of one vertex and one neighbouring vertex, since their position vectors are independent and orientation then fixes the image of their cross product. So that group has at most $4 \cdot 3, 6 \cdot 4$, or $12 \cdot 5$ elements; these are 12, 24, 60, which equal N . $\square$

Quiz 4.5. Check the pole count for D_n : list the orbits of poles and their stabilizer orders, and verify $2 - 2/2n = (1 - \frac{1}{2}) + (1 - \frac{1}{2}) + (1 - \frac{1}{n})$.

Chapter 10 imposes the preservation of a lattice, which admits only $n \in \{1, 2, 3, 4, 6\}$ and excludes the icosahedral group, and adjoins the improper operations, to obtain the 32 crystallographic point groups.

4.7. INDUCED ACTION ON FUNCTION SPACES

READING. [Moore, §5.5, pp. 54–58].

An action on a set induces an action on the functions defined on that set.

induced action
诱导作用

Definition 4.36 (Induced action). Let X be a G -set and Y a nonempty set, and let $F[X \rightarrow Y]$ be the set of maps $f : X \rightarrow Y$. For $g \in G$ and $f \in F[X \rightarrow Y]$ define $g \cdot f \in F[X \rightarrow Y]$ by

$$(g \cdot f)(x) := f(g^{-1} \cdot x).$$

Proposition 4.37. *This is an action of G on $F[X \rightarrow Y]$. If Y is a vector space, each g acts linearly. If moreover X is finite and $Y = \mathbb{C}$, then $F[X \rightarrow \mathbb{C}]$ is a vector space of dimension $|X|$ with basis the delta functions δ_x , where $\delta_x(y) = 1$ if $y = x$ and 0 otherwise, the action is a homomorphism $G \rightarrow GL(F[X \rightarrow \mathbb{C}])$, and*

$$g \cdot \delta_x = \delta_{g \cdot x}.$$

Proof. $(e \cdot f)(x) = f(x)$, and for two elements

$$(g_1 \cdot (g_2 \cdot f))(x) = (g_2 \cdot f)(g_1^{-1} \cdot x) = f(g_2^{-1} \cdot (g_1^{-1} \cdot x)) = f((g_1 g_2)^{-1} \cdot x) = ((g_1 g_2) \cdot f)(x),$$

using $g_2^{-1} g_1^{-1} = (g_1 g_2)^{-1}$. When Y is a vector space, $g \cdot (af_1 + bf_2) = a(g \cdot f_1) + b(g \cdot f_2)$ holds pointwise. For finite X and $Y = \mathbb{C}$, every f is $\sum_x f(x) \delta_x$ uniquely, so the δ_x are a basis; each g is then an invertible linear map, with inverse g^{-1} , hence an element of $GL(F[X \rightarrow \mathbb{C}])$ as in §3.5; and $(g \cdot \delta_x)(y) = \delta_x(g^{-1} \cdot y)$ is 1 exactly when $y = g \cdot x$. $\square$

The inverse in the definition is what makes the order come out right: with $f(g \cdot x)$ instead, the same computation produces $(g_2 g_1) \cdot f$.

For finite X and $Y = \mathbb{C}$, the matrix of g in the delta-function basis is the permutation matrix of its action on X . A function on the four vertices of the square is the ordered list $(f(\vec{v}_1), f(\vec{v}_2), f(\vec{v}_3), f(\vec{v}_4))$, so the induced action of D_4 on $\mathbb{C}^4$ is by 4×4 permutation matrices.

4. Group actions

Example 4.38 (Translating a function). Take $X = \mathbb{R}$ with the translation action $a \cdot x = x + a$ and $Y = \mathbb{C}$. Then

$$(T_a \psi)(x) = \psi(x - a).$$

Translating by a moves the graph of ψ to the right by a : the new function at x has the value the old one had at $x - a$.

Quiz 4.6. In the ordered basis $(\delta_{v_1}^+, \delta_{v_2}^+, \delta_{v_3}^+, \delta_{v_4}^+)$, write the 4×4 matrices of A and B from Example 4.7 and verify $BAB = A^{-1}$.

KEY POINTS

- An action of G on X is a homomorphism $G \rightarrow S_X$; orbits partition X and stabilizers are subgroups.
- Cosets are the orbits of H acting by $h \cdot g = gh^{-1}$, each in bijection with H . For finite G their partition gives Lagrange's theorem, $|G| = [G : H] |H|$. Orbit-stabilizer identifies $G \cdot x \cong G/G_x$, so $|G \cdot x| = [G : G_x]$.
- Conjugacy classes are the orbits of G on itself by conjugation, of size $[G : C_G(h)]$. The class equation gives a nontrivial centre to every nontrivial p -group; cyclic shifts of tuples prove Cauchy's theorem.
- The finite subgroups of $SO(3)$ are C_n , D_n , and the three polyhedral groups, by counting poles and then identifying the solids.
- An action on X induces an action on functions on X by $(g \cdot f)(x) = f(g^{-1}x)$, linear when the values lie in a vector space; the inverse makes it a left action.

KEY TERMS. action 群作用 · orbit 轨道 · stabilizer 稳定化子 · fixed point 不动点 · coset 陪集 · index 指数 · conjugacy class 共轭类 · centralizer 中心化子 · normalizer 正规化子 · class function 类函数 · class equation 类方程

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